Ryan Karami

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Education

Georgia Institute of Technology | Atlanta, GA, USA May 2028
Bachelor of Science in Electrical Engineering, GPA 4.00/4.00

Skills

Technical: RTL development, Verification, Static Timing Analysis (STA), Computer Architecture, Clock Domain Crossing (CDC)
Hardware: AMD FPGA (Spartan 7), Altera Cyclone-V FPGA, ESP32, Arduino, Teensy 4.1, oscilloscope, logic analyzer
Programming: C / C++, SystemVerilog, VHDL, MATLAB, RISC-V assembly, Git, Python, Linux
Software: AMD Vivado, Altera Quartus, Modelsim, Cadence Innovus, Altium Designer, GitHub
Languages: English (fluent), French (proficient)

Projects

Programmable SPI | SiliconJackets | *(RTL, SystemVerilog)* August 2026 – October 2026
Created a programmable Serial Peripheral Interface master communication module for a larger RISC-V processor.
- Implemented 2 synchronous FIFOs and memory mapped registers increasing CPU to SPI master interfacing by 80%.
- Optimized SystemVerilog RTL for area, achieving 5% reduction from initial implementation.
- Conducted preliminary test-benching, achieving 100% functional verification of module.

RISC-V CPU | Personal | *(AMD Xilinx FPGA, RTL, Computer Architecture, CDC)* June 2026 – August 2026
Worked in a team of 3 to architect a 5-stage pipelined RV32I processor on an AMD Spartan-7 FPGA.
- Created 32KB Harvard architecture instruction and data caches, 2-bit dynamic branch prediction, 256MB DDR3 memory integration, AXI-interconnects, and HDMI video output.
- Optimized SystemVerilog design RTL, achieving 90MHz CPU clock speed.
- Automated program boot-loading using a make-file and Python scripts to communicate with FPGA over UART.

FPGA FIR Filter | Academic | *(C++, Vitis HLS, Verilog)* November 2026
Created a 5-9-5 fir nulling filter on FPGA fabric to eliminate high frequencies in audio samples.
- Implemented a UART module in SystemVerilog to send audio sample data to FPGA at sampling rate of 8000Hz.
- Utilized C++ HLS to develop filter module running on a 100MHz clock and incorporated DSP slices for multiplication.

64 bit integer adder | SiliconJackets | *(Computer Arithmetic, UVM, STA)* January 2026 – February 2026
Designed a 2-cycle 64-bit adder utilizing a 32-bit addition core and dual SRAM interfaces to process large numerical datasets.
- Implemented SystemVerilog RTL and completed UVM verification reaching 98% IMC coverage.
- Executed physical design flow in Cadence Innovus targeting a 130nm process node.

Accessible Gaming Headband | Hackathon | *(C++, ESP32, Altium Designer)* March 2026
Collaborated in team of 4 to develop an accessible gaming headband for those with auditory disabilities.
- Developed circuitry and firmware that converts spatial audio into haptic feedback using 8 vibration motors and ESP32.
- Incorporated IMU and wrote C++ firmware to allow for head-tilt in game movement control.

Charge Control Unit Display | HyTech Racing | *(C++, Teensy 4.1)* January 2026
Upgraded the Charge Control Unit cart display by adding interactive functionality.
- Programmed button press capabilities in C++ to toggle screens on vehicle's Charger Control Unit display.
- Unpacked CAN messages, displaying 120+ battery cell temperatures and voltages ensuring normal charging conditions.

Work Experience

Nallapaneni Consultants LLC | Richmond, VA May 2026 – July 2026
Electrical Engineering Intern
Structural Engineering firm focusing on tunnels and movable bridge design and inspections.
- Inspected tunnel electrical systems ensuring electrical safety standards in $3.9 Billion state reconstruction project.
- Created Excel and VBA automation tools to handle tunnel electrical design calculation procedures.
- Developed automated Python data acquisition and processing pipelines for 20+ bridge Structural Health Monitoring systems.
- Conducted preliminary review of tunnel and movable bridge electrical and logic control system schematics.

Relevant Coursework

Digital Design; Signal Processing; Hardware/Software Programming; Circuit Analysis